

Estimating Relatedness in Admixed and Structured Populations

An overview of existing methods

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1 The problem

Classical relatedness estimators — the method-of-moments estimator in PLINK, or maximum-likelihood estimation of the Cockerham coefficients (k_0, k_1, k_2) — assume that both individuals are drawn from a single, homogeneous population with known allele frequencies p . Under admixture this assumption fails in two ways:

- **Population-level allele frequencies are the wrong frequencies.** Two unrelated individuals who share ancestry from the same source population share alleles more often than the pooled frequency predicts. They are scored as related; kinship is inflated, and the inflation scales with how differentiated the source populations are (F_{ST}).
- **Frequencies are individual-specific.** For an individual with ancestry proportions $q_i = (q_{i1}, \dots, q_{iK})$ and ancestry-specific frequencies $f_1, \dots, f_K$, the relevant frequency is $\pi_i = \sum_k q_{ik} f_k$, which differs between the two members of a pair. A pair with *different* ancestry profiles is then scored as *less* related than unrelated — KING-robust, for instance, returns kinship estimates around -0.13 for unrelated pairs of dissimilar ancestry.

Essentially all methods below are answers to the same question: *what do you condition on so that “sharing alleles” means “sharing recent ancestors” rather than “sharing continental ancestry”?* Three families of answers exist, and they are summarised in Table 1.

2 Three strategies

2.1 (A) Condition on estimated ancestry: ancestry-specific allele frequencies

Run a global ancestry / admixture model first (ADMIXTURE, STRUCTURE, frappe, NGSadmix) to obtain $\hat{q}_i$ and $\hat{f}_k$, then plug the individual-specific frequencies $\hat{\pi}_i = \sum_k \hat{q}_{ik} \hat{f}_k$ into a relatedness model. This is the most explicit approach and gives interpretable IBD-state output, at the cost of depending on K being sensible and on the admixture model fitting.

- **REAP** (Thornton et al. 2012) — moment estimator of (k_0, k_1, k_2) and kinship using individual-specific frequencies conditional on genome-wide ancestry. The first method to make this idea standard; validated under ancestry-related assortative and disassortative mating.
- **RelateAdmix** (Moltke & Albrechtsen 2014) — full maximum-likelihood estimation of (k_0, k_1, k_2) given $\hat{q}$ and $\hat{f}$, accounting for the fact that the two IBD alleles must come from the *same* source population. R package plus a multithreaded C++ command-line tool.
- **InRelate** (Sethuraman 2018) — ML over the full nine-state Jacquard IBD framework (so inbred as well as outbred pairs), for multi-allelic markers, given admixture-model ancestry.
- **SEEKIN** (Dou et al. 2017) — kinship for shallow / off-target sequencing and array data; models genotype uncertainty (dosages from imputation) and corrects for structure using ancestry coordinates. Works down to $\sim 0.15 \times$ depth; **SEEKIN-het** is the admixture-aware mode.
- **NGSremix** (Nøhr et al. 2021) — ML estimation of (k_0, k_1, k_2) for admixed pairs from *genotype likelihoods*, so genotypes are never called. Designed for low-depth NGS ($\leq 4\times$), where calling genotypes would itself bias relatedness; outperforms PLINK, KING, RelateAdmix and NgsRelate in that regime.

2.2 (B) Condition on principal components: PC-derived individual-specific frequencies

Instead of a discrete K -population model, regress genotypes on ancestry-representative PCs to predict each individual’s expected allele frequency, and define relatedness from the residuals. No choice of K , no assumption that ancestral populations are discrete — this is the “model-free” route.

- **PC-AiR** (Conomos, Miller & Thornton 2015) — *not* a relatedness estimator, but the necessary first half. Ordinary PCA in a sample containing relatives captures family structure rather than ancestry. PC-AiR partitions the sample into an ancestry-representative unrelated subset, runs PCA there, and projects the relatives back, giving PCs that reflect ancestry and not pedigree.

- **PC-Relate** (Conomos, Reiner, Weir & Thornton 2016) — the second half. Uses the PC-AiR PCs to obtain individual-specific allele frequencies, then estimates kinship coefficients, IBD-sharing probabilities and inbreeding coefficients from the residual genotype correlations.
- In practice the two are run as one iterative pipeline (PC-AiR → PC-Relate → refined unrelated set → PC-AiR again), implemented in the Bioconductor package **GENESIS**. This is the de-facto standard in large human cohorts (TOPMed, UK Biobank-style analyses).

2.3 Where the individual allele frequencies come from: PCAngsd and EMU

Both strategy (A) and strategy (B) reduce to the same quantity: the *individual allele frequency* π_{is} , the expected frequency at site s for individual i given their ancestry. Strategy (A) gets it from a discrete K -population decomposition, $\pi_{is} = \sum_k q_{ik} f_{ks}$; strategy (B) gets it from a low-rank (PCA) decomposition of the genotype matrix. Two tools from the low-rank side deserve separate mention, because they are what make the whole framework usable on data where genotypes cannot be trusted.

PCAngsd (Meisner & Albrechtsen 2018) estimates individual allele frequencies for *low-depth* NGS data directly from genotype likelihoods, so no genotype is ever called. The model is $\Pi = \mathbf{SA}$: individual frequencies are a low-rank product of a population-structure matrix and a loading matrix. It is fitted by an iterative EM-like scheme:

1. estimate population frequencies $\hat{p}_s$ from the genotype likelihoods;
2. form posterior genotype probabilities $P(G_{is} = g \mid X_{is}, \hat{\pi}_{is})$ by Bayes' rule and take *dosages* $E[G_{is} \mid X_{is}, \hat{\pi}_{is}]$;
3. centre the dosages, truncate their SVD at D components, and add $2\hat{p}_s$ back to obtain $\hat{\Pi}$;
4. feed $\hat{\pi}_{is}$ back in as the prior for step 2, and iterate to convergence.

The covariance matrix is then computed from dosages centred on $2\hat{p}_s$, which is valid under structure precisely *because* the dosages already condition on $\hat{\pi}$. The same $\hat{\Pi}$ is then reused downstream: admixture proportions by NMF, $\hat{\Pi} \approx QF^T$ (`--admix`), which is exactly the $\hat{q}, \hat{f}$ that RelateAdmix and NGSremix need, plus inbreeding coefficients, HWE tests, selection scans and genotype calling. In other words, PCAngsd supplies the input for strategy (A) as well as the PCs of strategy (B) from a single fit.

One caveat, since it is natural to assume otherwise: **PCAngsd does not itself estimate pairwise kinship any more**. Versions up to 0.99 had a `-kinship` flag implementing a PC-Relate-style (Conomos) kinship matrix adapted to genotype likelihoods, plus a `-relate` option for filtering related individuals out of a Beagle file. Both were dropped in the January 2021 Cython rework, and the current release (v1.36) exposes only *inbreeding* estimators (`--inbreed-sites`, `--inbreed-samples`). PCAone is the same: `--inbreed`, but no kinship. In a present-day pipeline PCAngsd and PCAone are therefore the *front-end* producing $\hat{\pi}$ or $\hat{q}, \hat{f}$; the relatedness estimate itself comes from NGSremix, RelateAdmix or PC-Relate.

EMU (Meisner, Liu, Huang & Albrechtsen 2021) solves the neighbouring problem: PCA when the data are *called* genotypes but enormous fractions are *missing* — ultra-low-coverage sequencing, NIPT data, merged arrays with different marker sets, ancient DNA. Standard PCA has to impute missing entries (usually with the mean), and individuals then separate along PCs according to *how much data they are missing* rather than their ancestry. EMU uses an accelerated EM algorithm for PCA that models the missingness explicitly and iteratively, alternating between a truncated SVD and re-imputation of the missing entries from the current low-rank fit — structurally the same idea as PCAngsd's loop, with missingness in place of genotype uncertainty. It scales to biobank-sized data (Python implementation).

PCAone (Li, Meisner & Albrechtsen 2023) is the scalable implementation of this whole layer. It is a C++ PCA framework offering three fast SVD algorithms — implicitly restarted Arnoldi (`--svd 0`), single-pass randomized SVD with power iterations (`--svd 1`) and the proposed window-based randomized SVD, winSVD (`--svd 2`) — each with in-core *and* out-of-core modes, so biobank-scale matrices need not fit in RAM. Crucially for the present purpose it re-implements *both* the EMU and the PCAngsd algorithms, so missingness and genotype-likelihood uncertainty are handled inside the same fast framework, reading PLINK BED, PLINK2 PGEN, BGEN, Beagle and CSV. It also provides ancestry-adjusted LD statistics for pruning and clumping — relevant here because LD pruning with *unadjusted* LD in a structured sample preferentially removes exactly the ancestry-informative sites the relatedness estimator needs to be conditioned on.

For relatedness work the practical point is the choice of front-end:

- *low depth, genotype likelihoods available* → PCAngsd (or NGSadmix for discrete K), then NGSremix for the actual relatedness estimation;
- *called genotypes, heavy and non-random missingness* → EMU for the ancestry axes, then a PC-based relatedness estimator;

- *called genotypes, complete data* $\rightarrow$ PC-AiR, or ADMIXTURE for discrete K .

A relatedness estimate is only as good as the $\hat{\pi}$ it conditions on, so this step is not a preliminary — it is half the method.

2.4 Correlation of residuals as a kinship estimator: evalAdmix

There is a fourth route into the same quantity, arrived at from the opposite direction. **evalAdmix** (Garcia-Erill & Albrechtsen 2020) was built to *test the fit* of an admixture analysis, not to estimate relatedness. Given genotypes and the output of ADMIXTURE/STRUCTURE/NGSAdmix, it forms the residual $g_{is} - 2\hat{\pi}_{is}$ between the observed genotype and the genotype predicted by the admixture model, and reports the matrix of pairwise *correlations of residuals*. Under a correctly specified model the residuals of two individuals are uncorrelated, so entries near zero indicate good fit; systematic positive correlation within groups and negative correlation between groups indicates the model is wrong (too small a K , ghost populations, drift after admixture, non-discrete ancestral populations).

But note what the residual is: exactly the deviation from the individual allele frequency. Two individuals who share recent ancestors deviate from $\hat{\pi}$ in the *same direction*, so relatedness produces a positive residual correlation — which is precisely PC-Relate’s estimator, with the admixture model in place of the PC regression. The correlation of residuals is therefore a kinship estimate; its documentation flags this the other way round, warning that a positive correlation “might also be due to relatedness”. Concretely, the residual correlation estimates $r_{ij} = 2\phi_{ij}$, the coancestry/relatedness coefficient, not ϕ_{ij} itself.

This is easy to confirm. Running evalAdmix on the 126-individual example data distributed with RelateAdmix ($K = 2$, 104,290 SNPs) and comparing against RelateAdmix’s own ML estimates on the same data and same $\hat{q}, \hat{f}$:

Relationship class	$\hat{\phi}$ (RelateAdmix)	mean corr. of residuals	n pairs
unrelated	0.000	−0.003	7816
second cousins	0.012	0.036	9
first cousins	0.053	0.124	10
half sibs	0.111	0.245	10
PO / full sibs	0.247	0.494	20
MZ / duplicates	0.500	1.000	10

The Pearson correlation between the two is $r = 0.988$, and the regression is $\text{corr. resid.} = 2.02\hat{\phi} - 0.004$ — the factor of two expected if the residual correlation estimates 2ϕ . Duplicate pairs return exactly 1.000. Unrelated pairs centre on -0.003 with SD 0.007.

Two practical consequences. First, if you are running evalAdmix to choose K , close relatives will show up as bad model fit and can be mistaken for a K problem — so remove relatives first, or read the outliers as relatedness. Second, in the other direction, an evalAdmix run you already have is a usable relatedness screen for free.

The PCA version (van Waaij, Li, Garcia-Erill, Albrechtsen & Wiuf 2023) generalises this to population structure inferred by *either* PCA or the admixture model, and replaces the original leave-one-out frequency correction with an analytic bias correction: compare the empirical correlation of residuals $\hat{b}$ with the correlation $\hat{c}$ implied by the model itself, obtained from $(\mathbf{I} - \mathbf{P})\hat{\mathbf{D}}(\mathbf{I} - \mathbf{P})$ where $\mathbf{P}$ projects onto the column space of Q (or of the PCs). Under a correct model with $k' = k$, $\hat{b} - \hat{c} \rightarrow 0$ as the number of SNPs grows. It is single-pass rather than iterative, needs only genotypes and Q (no ancestral frequency file), and is available in evalAdmix v1.0 as `-method corrected`, with an R implementation at github.com/popgenDK/evalPopStructure. Tellingly, in their 1000 Genomes analysis the elevated corrected coefficients are explained by five pairs of first-degree relatives — i.e. the statistic is picking up kinship exactly as above.

2.5 (C) Avoid allele frequencies altogether

Use only the joint genotype-sharing pattern within the pair, so that no external frequency — correct or incorrect — can bias the estimate.

- **KING-robust** (Manichaikul et al. 2010) — kinship from counts of shared heterozygotes versus opposing homozygotes. Robust to *discrete* structure and to small samples, and universally used as a fast screen. Its documented failure mode is exactly the admixture case: unbiased when both members of a pair have similar admixture proportions, systematically negative when they do not.

Table 1: Methods for relatedness estimation under admixture / population structure.

Method	Strategy	Input data	Ancestry handled via	Output / notes
PLINK --genome	baseline	called genotypes	<i>nothing</i>	(k_0, k_1, k_2) ; biased under admixture
NgsRelate	baseline	genotype likelihoods	<i>nothing</i>	(k_0, k_1, k_2) ; single-population, low depth
KING-robust	(C)	called genotypes	pairwise genotype sharing	kinship; fast screen, biased for pairs of <i>differing</i> ancestry
R_0/R_1 /KING	(C)	genotypes or GLs	none needed	relationship class; no allele freqs, no positions
REAP	(A)	called genotypes	$\hat{q}, \hat{f}$ from ADMIXTURE etc.	(k_0, k_1, k_2) , kinship; moment estimator
RelateAdmix	(A)	PLINK bed + .P/.Q	$\hat{q}, \hat{f}$ from ADMIXTURE etc.	(k_0, k_1, k_2) ; ML, multithreaded C++ / R
InRelate	(A)	multi-allelic genotypes	$\hat{q}, \hat{f}$	all 9 Jacquard states; handles inbreeding
SEEKIN	(A)	sparse seq. / dosages	ancestry coordinates + ref. panel	kinship; works at $\sim 0.15 \times$ depth
NGSremix	(A)	genotype likelihoods	$\hat{q}, \hat{f}$ (NGSadmix/PCAngsd)	(k_0, k_1, k_2) ; best for low-depth admixed
PC-AiR	(B)	called genotypes	—	ancestry PCs unconfounded by relatedness
PC-Relate	(B)	called genotypes	PC-AiR PCs	kinship, IBD probs, inbreeding; “model-free”
PCAngsd	front-end	genotype likelihoods	iterative SVD on dosages $\rightarrow \hat{\pi}$	$\hat{\pi}$, PCA, $\hat{q}, \hat{f}$ (NMF), inbreeding; <i>no</i> kinship since v1.0
EMU	front-end	called genotypes, much missing	EM-PCA modelling missingness	ancestry PCs unbiased by missingness
evalAdmix	(A) residuals	PLINK bed or Beagle GLs	residual vs. $2\hat{\pi}$ from $\hat{q}, \hat{f}$	corr. of residuals $\approx 2\phi$; built as a model-fit test
evalAdmix -method corrected	(A)/(B) residuals	PLINK bed	analytic $\hat{b} - \hat{c}$ correction, Q or PCs	as above, single-pass, no .P needed
PCAone	front-end	BED/PGEN/BGEN/BGEM/Beagle/PCAngsd	EM-SVD/PCAngsd implements EMU & PCAngsd	fast in-/out-of-core PCA, $\hat{\pi}$, inbreeding, ancestry-adj. LD; no kinship

- R_0, R_1 , **KING-robust kinship** (Waples, Albrechtsen & Moltke 2019) — three genotype-sharing summary statistics, of which the pairs (R_1, R_0) and (R_1, KING) separate parent–offspring, full sibs, half sibs/avuncular/grandparent and unrelated without any allele frequencies and without genomic positions. Robust to ascertainment bias; well suited to non-model organisms, ancient DNA and reference-free settings. Distinguishes relationship *classes* rather than returning unbiased (k_0, k_1, k_2) .

3 From kinship to (k_0, k_1, k_2) : what PC-Relate actually computes

One clarification first: **PC-AiR does not do this**. PC-AiR produces only the ancestry PCs; the IBD sharing probabilities come from **PC-Relate**. The description below follows the GENESIS implementation (`pcrelate`, `ibd.probs = TRUE`), where $g_{is} \in \{0, 1, 2\}$ is the allele count and $\hat{\mu}_{is}$ is the individual-specific allele frequency obtained by regressing genotypes on the PC-AiR PCs (fitted on a training set of mutually unrelated individuals) and halving the prediction.

Kinship. Form the residual $R_{is} = g_{is} - 2\hat{\mu}_{is}$ and take a normalised cross-product:

$$\hat{\phi}_{ij} = \frac{\sum_s R_{is} R_{js}}{4 \sum_s \sqrt{\hat{\mu}_{is}(1 - \hat{\mu}_{is}) \hat{\mu}_{js}(1 - \hat{\mu}_{js})}}.$$

k_2 from a dominance coding. Define the dominance-coded genotype

$$D_{is} = \begin{cases} \hat{\mu}_{is} & g_{is} = 0 \\ 0 & g_{is} = 1 \\ 1 - \hat{\mu}_{is} & g_{is} = 2 \end{cases} \quad \tilde{D}_{is} = D_{is} - \hat{\mu}_{is}(1 - \hat{\mu}_{is}),$$

which has $E[\tilde{D}_{is}] = 0$ under HWE. Its cross-product isolates the IBD-2 component, because the dominance covariance between two individuals is proportional to k_2 alone:

$$\hat{k}_2 = \frac{\sum_s \tilde{D}_{is} \tilde{D}_{js}}{\sum_s \hat{\mu}_{is}(1 - \hat{\mu}_{is}) \hat{\mu}_{js}(1 - \hat{\mu}_{js})}.$$

k_0 from opposing homozygotes. If a pair shares *any* allele IBD at a site, they cannot be opposing homozygotes. So the observed IBS0 count, divided by the count expected if the pair shared nothing, estimates k_0 directly:

$$\hat{k}_0 = \frac{\sum_s [\mathbf{1}(g_{is}=2, g_{js}=0) + \mathbf{1}(g_{is}=0, g_{js}=2)]}{\sum_s [\hat{\mu}_{is}^2(1 - \hat{\mu}_{js})^2 + (1 - \hat{\mu}_{is})^2 \hat{\mu}_{js}^2]}.$$

k_1 by subtraction. $\hat{k}_1 = 1 - \hat{k}_0 - \hat{k}_2$. GENESIS stores **kin**, **k0** and **k2**; k_1 is left as the remainder.

So the direct answer to “how does it go from kinship to IBD sharing” is: *for close pairs, it does not*. $\hat{\phi}$, $\hat{k}_0$ and $\hat{k}_2$ are three separate estimators computed from the same residuals, and the identity $\phi = k_1/4 + k_2/2$ is not used to derive them. That matters because parent–offspring and full sibs both have $\phi = 1/4$ — kinship alone cannot tell them apart, and it is the IBS0-based $\hat{k}_0$ (0 versus $1/4$) that does.

The corrections, applied only when IBD probabilities are requested. Three, in order:

- *Inbreeding.* Departure from HWE inflates the dominance covariance, so PC-Relate subtracts the product of the two individuals’ inbreeding coefficients (the diagonal of the same output): $\hat{k}_2 \leftarrow \hat{k}_2 - \hat{f}_i \hat{f}_j$.
- *Residual ancestry and kinship bias in $\hat{k}_2$ (small-sample correction).* For each PC $k \geq 2$, regress the current $\hat{k}_2$ on the entries of $V_k V_k^\top$ with a linear and a quadratic term, fitting on pairs with $\hat{\phi} < 2^{-11/2} \approx 0.022$ (effectively unrelated), and subtract the fitted value. Then regress on $\hat{\phi}$ itself, fitting on pairs with $\hat{k}_2 < 2^{-9/2} \approx 0.044$, and subtract that too.
- *k_0 for more distant pairs.* Here the kinship identity finally enters. For pairs below the first-degree boundary, $\hat{\phi} < 2^{-5/2} \approx 0.177$, the IBS0 estimate is discarded and replaced by

$$\hat{k}_0 = 1 - 4\hat{\phi} + \hat{k}_2,$$

which is exactly $\phi = k_1/4 + k_2/2$ with $k_1 = 1 - k_0 - k_2$, rearranged. The reason for the switch is that in distant pairs the IBS0 count is dominated by noise and genotyping error, while $\hat{\phi}$ is still well estimated; in first-degree pairs the IBS0 estimator is kept, for the reason above.

4 Choosing a method

- **Array or high-depth genotypes, human cohort, cryptic structure:** PC-AiR + PC-Relate (GENESIS). No K to choose, and it is what large consortia use. Screen first with KING-robust.
- **You have a defensible admixture model** (a clear K , good reference panels): REAP or RelateAdmix. RelateAdmix gives proper ML (k_0, k_1, k_2), which matters if you want to distinguish full sibs from parent–offspring rather than just rank kinship.
- **Low-depth sequencing:** do not call genotypes. NGSremix (with NGSadmixmap or PCAngsd for $\hat{q}, \hat{f}$), or SEEKIN for very sparse/off-target data.
- **Non-model organism, ancient DNA, no reference frequencies:** R_0/R_1 /KING, which needs neither frequencies nor a genetic map.
- **Always:** report which ancestry model was used and at what K . Under strategy (A), the relatedness estimates are conditional on $\hat{q}$ and $\hat{f}$; a poorly chosen K propagates directly into the kinship estimates.

5 References

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